



Luis Felipe Rodriguez Rico
Python Developer — Data & Process Optimization
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Professional Profile

Development and optimization of Python scripts and pipelines focused on data processing, workflow automation, and legacy code refactoring in production environments. With a background in Data Analytics, my career evolved toward data engineering, backend development with Python APIs and frameworks, and critical server-side process optimization. I specialize in turning inefficient processes into robust and scalable solutions.

Technical Skills

Languages	Python, SQL, Dart, TypeScript, C#, C++, Java
Frameworks/Lib	FastAPI, Uvicorn, Pandas, NumPy, Flutter, Node.js, Vite
Databases	MySQL, SQLite, PostgreSQL, MongoDB (Atlas), MariaDB
Tools	Docker, Cloudflare, Git, REST APIs, Parallel Processing
Specialty	Query Optimization, ETL Pipelines, Backend, Automation

Professional Experience

Data Analyst & Backend Developer **NRTEC Desarrollos Tecnológicos**
2025 – Present

- Designed and implemented optimized data pipelines in **Python**, reducing analysis and comparison times by 70% through parallel processing, chunk-based ingestion, and automated debugging with **Pandas** and **NumPy**.
- Developed an internal web platform integrating a high-performance backend with **FastAPI** / **Uvicorn** and a frontend with **Node.js (Vite)**, automating extraction from multiple database schemas.
- Eliminated manual Excel-based reporting processes by creating a one-click automated generation and analysis system, accelerating operational decision-making.

Data Analyst (Hackathon) **Hackathon Magna**
2025

- Led the feature engineering phase under NDA, performing exploratory data analysis (EDA) on raw data, detecting complex patterns, and assigning statistical weights for predictive models.
- Designed and structured clean, normalized datasets in **Python**, optimizing the team’s workflow for immediate execution of prediction algorithms.

Contestant **ICPC — Competitive Programming**
2025

- Solved complex logic problems using advanced data structures and algorithmic optimization under high time pressure in collaborative environments.

Education

Computer Engineering **2021 – December 2026**
Universidad Autónoma de San Luis Potosí *Full-time availability*

Featured Projects

Smart Cafeteria System | github.com/ELIMSS50/BeanRush
Full-stack platform for real-time role management and orders, integrated with **MongoDB Atlas** persistence and automatic notifications via **Gmail API** (Angular, Node.js).

Self-Checkout Store Mobile App | github.com/ELIMSS50/Proyecto-Movil
Mobile commerce application with contactless payment system through scanning and dynamic QR code generation, connected to a scalable backend in **TypeScript (Flutter, Dart)**.

Languages

English B1 (Intermediate)